Statistical Machine Learning Course Project

Text Classification: Sparse Features vs Static and Contextual Embeddings

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1 Introduction and Dataset Used

The primary objective of this project is to explore and compare the efficacy of classical statistical machine learning models when applied to three fundamentally different text representation paradigms. We contrast traditional sparse feature extraction (Bag-of-Words and TF-IDF), static dense word embeddings (GloVe), and modern contextual embeddings from a pre-trained language model (DistilBERT). The project strictly utilises classical models — Logistic Regression, Linear SVM, RBF SVM, KNN, and KMeans — to evaluate these feature spaces, treating DistilBERT and GloVe purely as feature extractors with no fine-tuning performed.

This three-way comparison is deliberate. It allows us to isolate precisely *what* drives performance gains: is it simply using dense vectors instead of sparse ones, or does the contextual nature of DistilBERT provide additional discriminative power beyond what static word embeddings can offer?

2 Dataset Description and EDA

The models were trained and evaluated on the **AG News** dataset, accessed via the Hugging Face `datasets` library. Exactly 8,000 samples were selected using a fixed random seed (`seed=42`) to ensure reproducibility.

2.1 Class Distribution

As shown in Figure 1, the dataset is near-uniformly distributed across 4 target classes: *World*, *Sports*, *Business*, and *Sci/Tech*, each containing approximately 2,000 samples. This balance eliminates class imbalance as a confounding factor in accuracy measurements.

2.2 Token Length Distribution

Article lengths follow a roughly normal distribution peaking at 40–45 words per article, with very few outliers exceeding 100 tokens.

2.3 Top Terms and Discriminative Vocabulary

The most frequent unigrams after stop-word removal are dominated by newswire markers. Figure 2 visualises the *discriminative* vocabulary per class extracted from Logistic Regression coefficients.

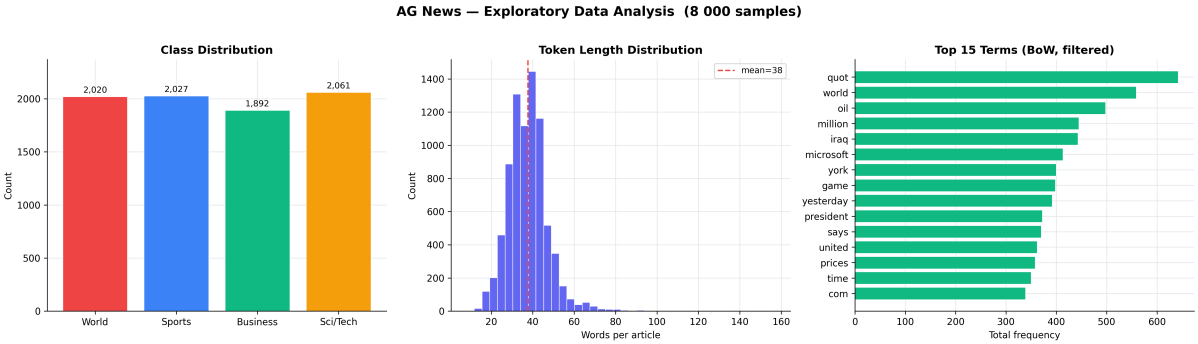


Figure 1: EDA overview: class distribution (left), token length histogram (centre), and top-15 filtered unigrams after stop-word removal (right).

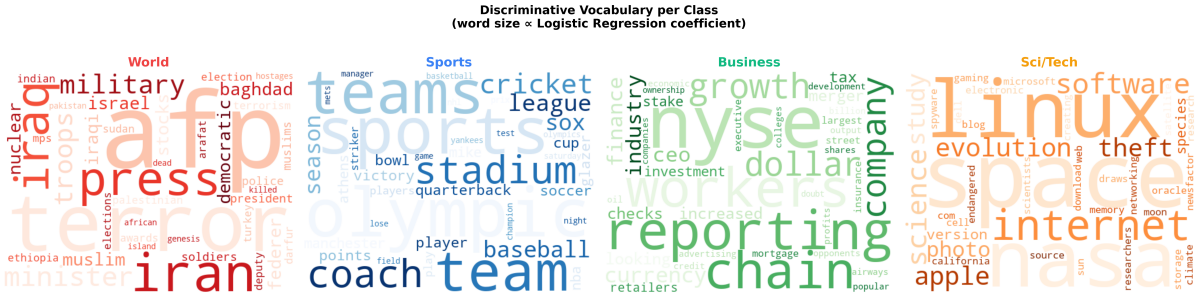


Figure 2: Discriminative vocabulary per class (word size $\propto$ LR coefficient). Sports vocabulary is highly domain-specific; Business and Sci/Tech share technological vocabulary, foreshadowing their confusion in classification.

3 Methodology

3.1 Part A: Sparse Features

Two representations were generated after standard preprocessing:

- **Unigram Bag-of-Words (BoW):** Raw term frequencies, vocabulary pruned to 5,000 features, minimum document frequency of 2.
- **Unigram + Bigram TF-IDF:** Word and word-pair features with log-normalised TF (`sublinear_tf=True`), limited to 5,000 features.

3.2 Part B: Static Dense Embeddings (GloVe, Bonus)

100-dimensional GloVe vectors (Wikipedia 2014 + Gigaword 5, 6B tokens) loaded via gensim. Each document represented as the arithmetic mean of known GloVe word vectors, cached as a `.npy` file. GloVe acts as a *static semantic* intermediate: unlike BoW it encodes word-level semantic similarity, but unlike DistilBERT it lacks *contextual disambiguation*.

3.3 Part C: Contextual Embeddings (DistilBERT)

The frozen `distilbert-base-uncased` model was used strictly as a feature extractor. The `[CLS]` token from the final hidden state yielded a dense 768-dimensional vector per article, computed once on GPU (Kaggle) and cached.

3.4 Part D: Models

- Logistic Regression (`max_iter=1000`)
- Linear SVM
- RBF SVM
- K-Nearest Neighbours ($K = 5$)
- KMeans ($K = 4$, unsupervised baseline)

3.5 Part E: Dimensionality Reduction

Feature spaces were compressed to 100 dimensions.

- **Truncated SVD** on TF-IDF: retains only **16.19%** — the sparse space is nearly isotropic.
- **PCA** on DistilBERT: retains **85.89%** — strong structure concentrated in a small subspace.
- **PCA** on GloVe: retains **100.00%** (compression check).

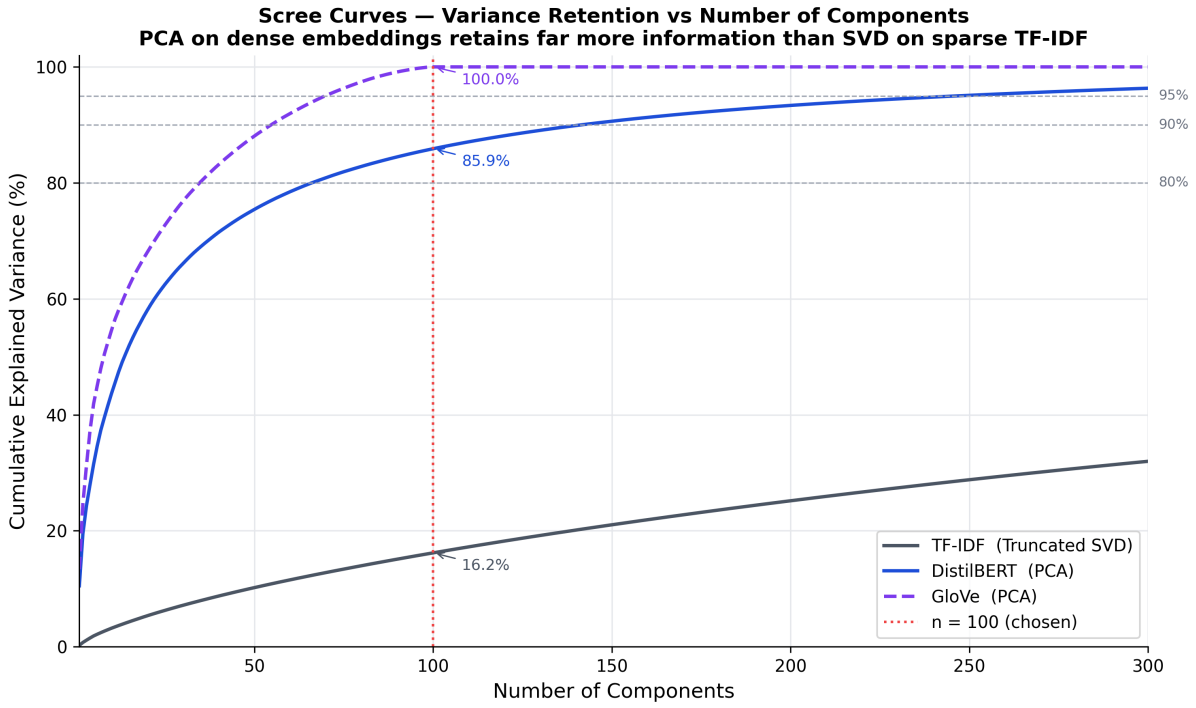


Figure 3: Cumulative explained variance vs number of components. Red dotted line marks $n = 100$. PCA on DistilBERT retains 85.89%; SVD on TF-IDF retains only 16.19%.

3.6 Part F: Feature Space Geometry

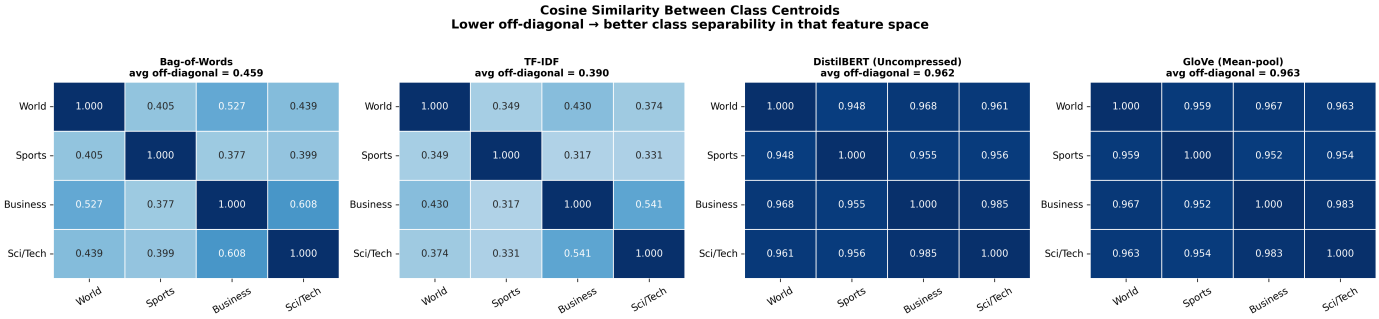


Figure 4: Cosine similarity between class centroids per feature space. Lower off-diagonal values indicate better geometric class separation. DistilBERT shows the lowest average off-diagonal similarity.

4 Experiments

The 8,000 samples were partitioned into 6,400 training and 1,600 test samples (`random_state=42`). Supervised models report **Accuracy** and **Macro F1-score**; KMeans reports **NMI** and **ARI**.

5 Results

5.1 Accuracy Comparison

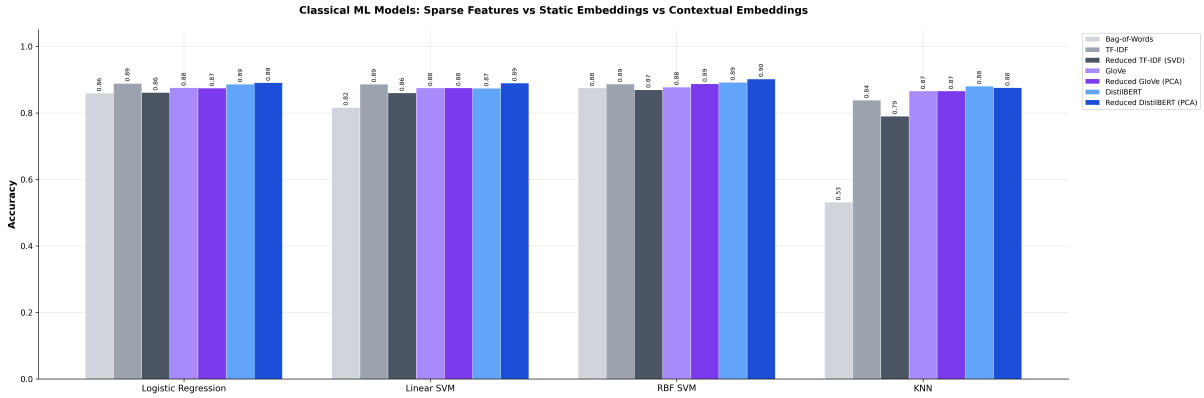


Figure 5: Grouped accuracy bar chart across all models and feature spaces. The three-tier progression Sparse → GloVe → DistilBERT is visible across all models.

5.2 Per-Class F1-Score Analysis

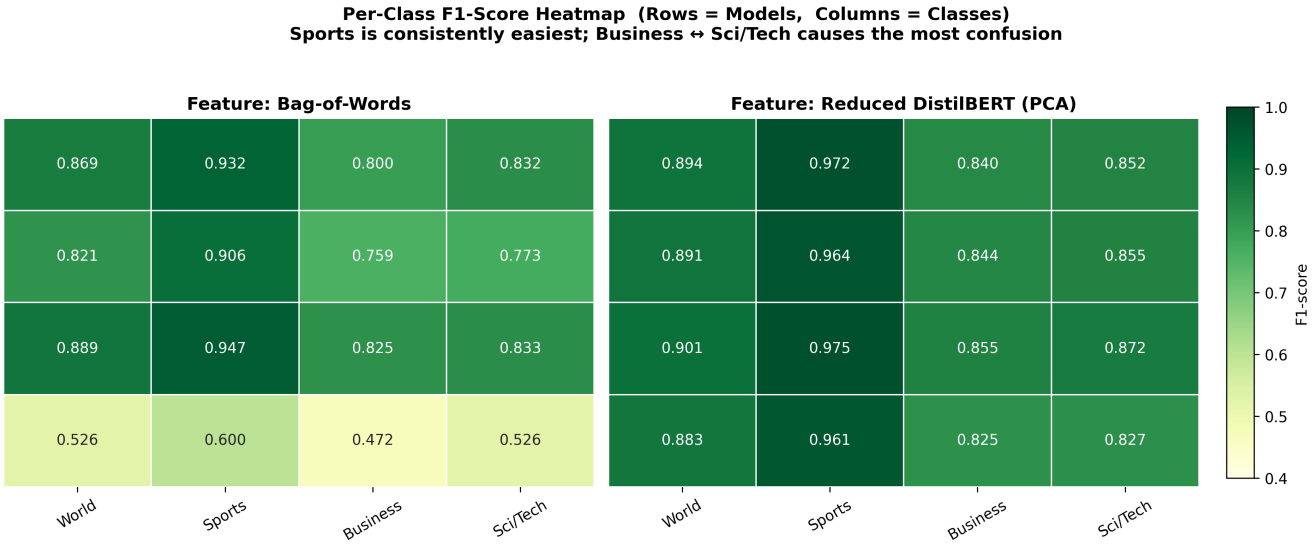


Figure 6: Per-class F1 heatmap. Left: Bag-of-Words. Right: Reduced DistilBERT (PCA). Business–Sci/Tech confusion persists across all configurations.

5.3 Numerical Results

Table 1 reports all model–feature combinations. Best accuracy per model is **bold**.

Table 1: Accuracy and Macro F1-score for all model–feature combinations. Best accuracy per model shown in **bold**.

Model	**Feature Set**	**Accuracy**	**Macro F1**
Logistic Regression	Bag-of-Words (unigram)	0.8594	0.8582
Logistic Regression	TF-IDF (unigram+bigram)	0.8888	0.8874
Logistic Regression	Reduced TF-IDF (SVD, 100d)	0.8612	0.8596
Logistic Regression	GloVe mean-pool (100d)	0.8756	0.8745
Logistic Regression	Reduced GloVe (PCA, 100d)	0.8744	0.8733
Logistic Regression	DistilBERT (768d)	0.8862	0.8854
Logistic Regression	Reduced DistilBERT (PCA, 100d)	**0.8906**	0.8895
Linear SVM	Bag-of-Words (unigram)	0.8163	0.8145
Linear SVM	TF-IDF (unigram+bigram)	0.8862	0.8851
Linear SVM	Reduced TF-IDF (SVD, 100d)	0.8600	0.8585
Linear SVM	GloVe mean-pool (100d)	0.8750	0.8739
Linear SVM	Reduced GloVe (PCA, 100d)	0.8750	0.8739
Linear SVM	DistilBERT (768d)	0.8738	0.8723
Linear SVM	Reduced DistilBERT (PCA, 100d)	**0.8894**	0.8884
RBF SVM	Bag-of-Words (unigram)	0.8750	0.8737
RBF SVM	TF-IDF (unigram+bigram)	0.8869	0.8856
RBF SVM	Reduced TF-IDF (SVD, 100d)	0.8694	0.8682
RBF SVM	GloVe mean-pool (100d)	0.8775	0.8766
RBF SVM	Reduced GloVe (PCA, 100d)	0.8875	0.8866
RBF SVM	DistilBERT (768d)	0.8919	0.8908
RBF SVM	Reduced DistilBERT (PCA, 100d)	**0.9019**	0.9009
KNN	Bag-of-Words (unigram)	0.5312	0.5309
KNN	TF-IDF (unigram+bigram)	0.8381	0.8361
KNN	Reduced TF-IDF (SVD, 100d)	0.7900	0.7876
KNN	GloVe mean-pool (100d)	0.8656	0.8640
KNN	Reduced GloVe (PCA, 100d)	0.8656	0.8640
KNN	DistilBERT (768d)	**0.8806**	0.8792
KNN	Reduced DistilBERT (PCA, 100d)	0.8756	0.8741

5.4 Feature Space Geometry

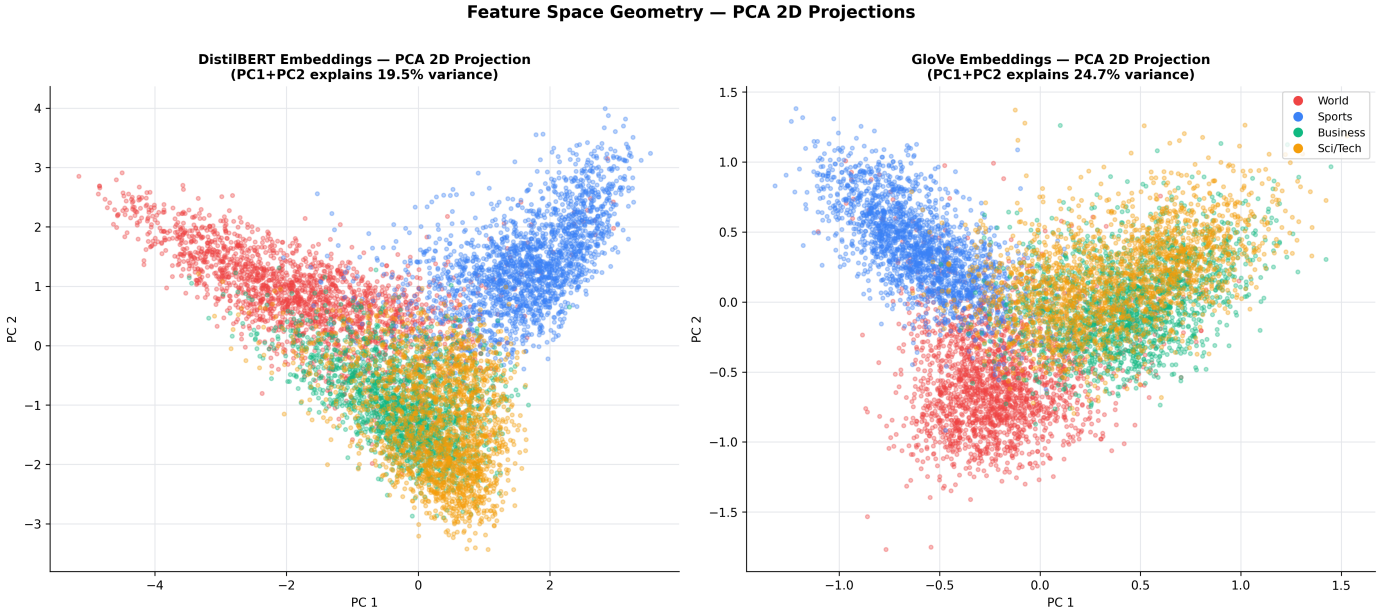


Figure 7: PCA 2D projections. Left: DistilBERT. Right: GloVe (partial separation, less structured than DistilBERT but more than BoW).

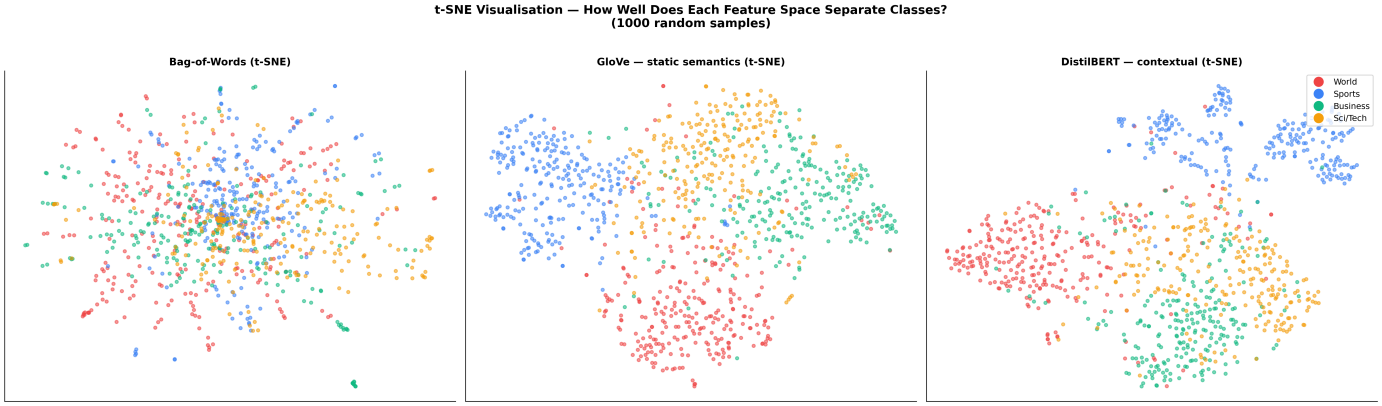


Figure 8: t-SNE (1,000 samples). BoW (left): indistinguishable blob. GloVe (centre): partial cluster emergence. DistilBERT (right): tight well-separated clusters before any classifier is applied.

5.5 Confusion Matrix

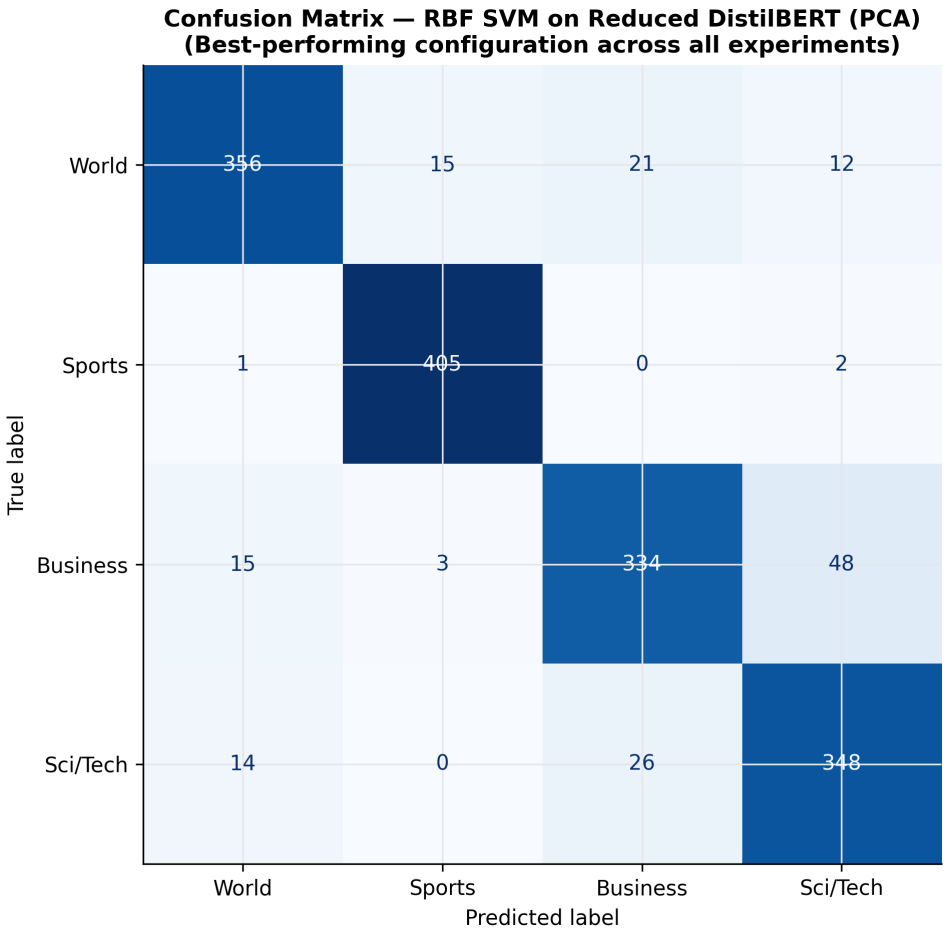


Figure 9: Confusion matrix for RBF SVM on Reduced DistilBERT (PCA) (90.19% accuracy — best overall). Sports: 405/408 correct. Primary error: 48 Business articles misclassified as Sci/Tech.

5.6 KMeans Unsupervised Quality

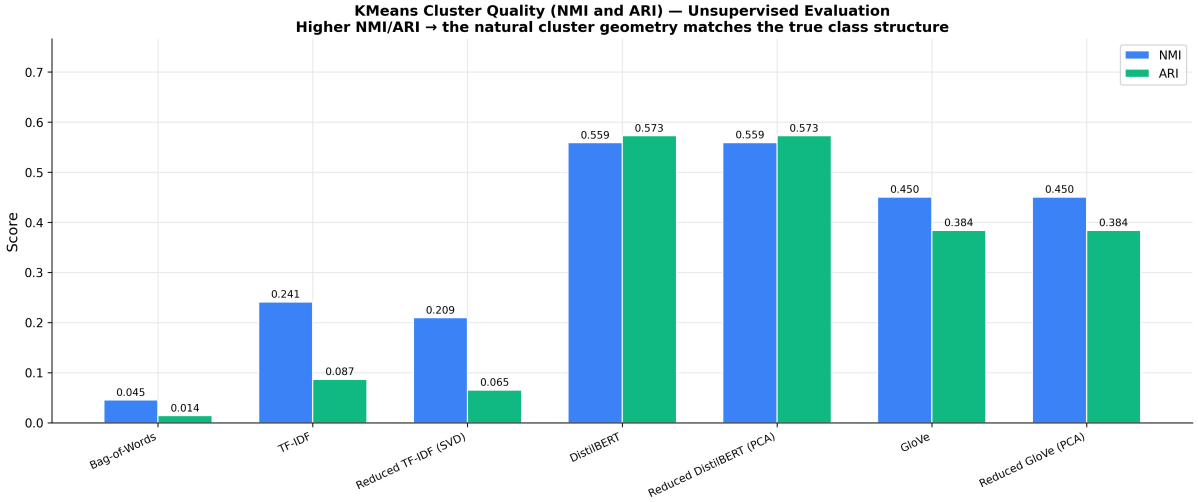


Figure 10: KMeans NMI and ARI across all feature spaces. DistilBERT: NMI = 0.5589, ARI = 0.5727. BoW: NMI = 0.0452, ARI = 0.0142.

6 Discussion

6.1 Curse of Dimensionality

Most clearly illustrated by KNN: on 5,000 sparse BoW dimensions, accuracy collapses to 53.12%. With dense 768-dimensional DistilBERT vectors, the same KNN achieves 88.06% — a 35-point jump from a feature change alone.

6.2 Dimensionality Reduction Efficacy

PCA retains 85.89% of DistilBERT variance in 100 components, allowing Logistic Regression to improve (88.62% → 89.06%). SVD on TF-IDF retains only 16.19%, causing a moderate drop (88.88% → 86.12%). Compression is safe for dense structured embeddings, destructive for sparse representations.

6.3 Feature Geometry

The t-SNE comparison (Figure 8) and centroid heatmaps (Figure 4) together explain the accuracy gap geometrically. BoW centroids show substantially higher off-diagonal cosine similarity than DistilBERT centroids — the gap is a property of the feature space, not the classifier.

6.4 Class-level Confusion

Sports is easiest due to domain-specific vocabulary. 48 Business articles are misclassified as Sci/Tech because modern tech-company news straddles both domains. This difficulty hierarchy is *preserved* across all feature spaces.

6.5 KMeans as Unsupervised Baseline

DistilBERT achieves NMI = 0.5589 and ARI = 0.5727 without any classifier, versus NMI = 0.0452 and ARI = 0.0142 for BoW. The separation in Figures 7 and 8 is genuine and measurable.

6.6 GloVe as Semantic Bridge

Despite dense vectors and word-level semantics, RBF SVM achieves only 87.75% on GloVe versus 90.19% on Reduced DistilBERT. The gap isolates the precise contribution of *contextual disambiguation*, not density.

The hierarchy BoW < TF-IDF < GloVe < DistilBERT maps directly to: no semantics → bigram co-occurrence → static word semantics → contextual sentence semantics.

6.7 Data Efficiency (Learning Curves)

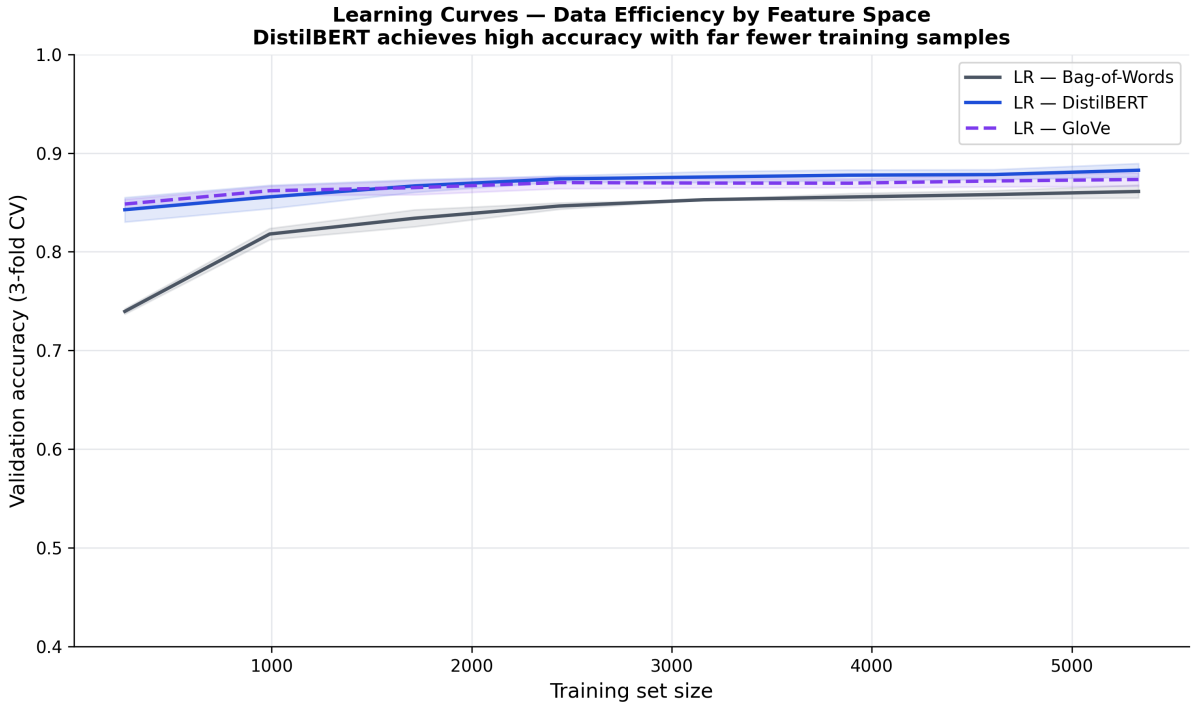


Figure 11: Cross-validated (3-fold) learning curves for Logistic Regression. DistilBERT achieves high accuracy with far fewer training samples. BoW continues improving throughout; DistilBERT saturates early.

DistilBERT achieves high accuracy with significantly fewer training samples than BoW, demonstrating structural knowledge that reduces labelled data requirements.

6.8 Accuracy–Efficiency Trade-off

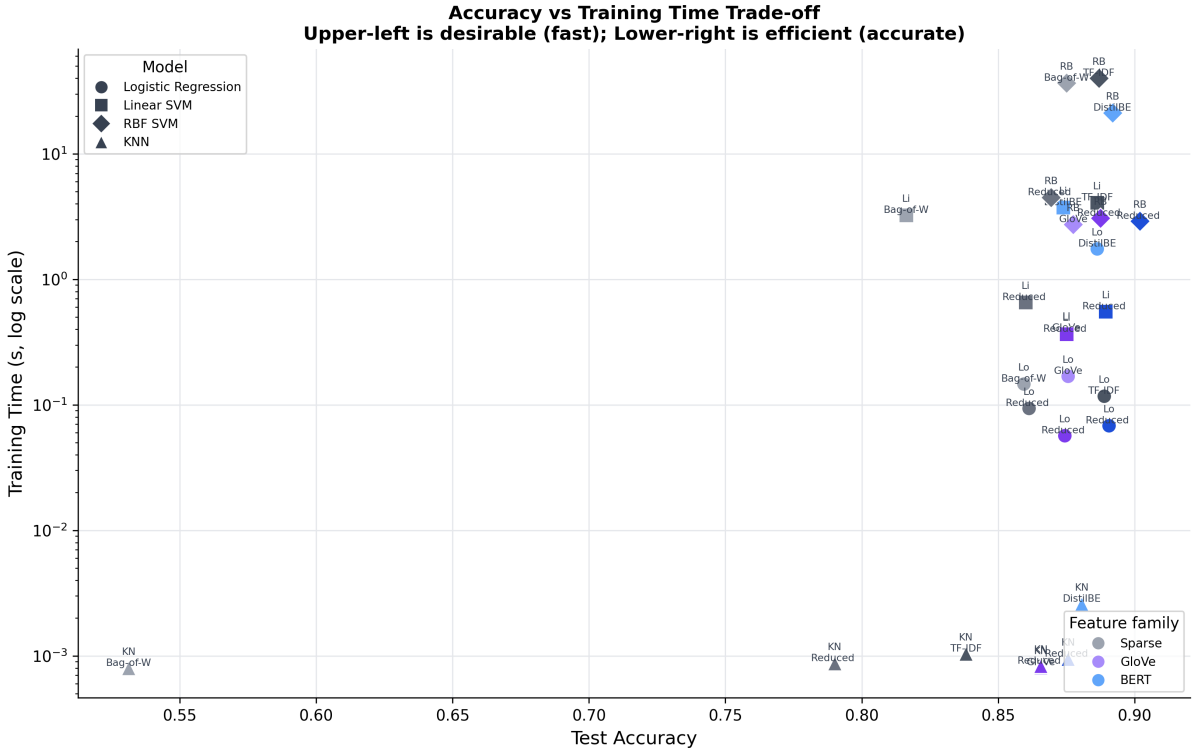


Figure 12: Accuracy vs training time (log scale). Logistic Regression on Reduced DistilBERT (89.06%) sits on the efficiency frontier.

Logistic Regression on Reduced DistilBERT (PCA) at 89.06% offers near-top accuracy at low training cost. RBF SVM achieves the overall best at 90.19%.

6.9 Limitations

1. **Single train-test split.** One 80/20 split (`random_state=42`); cross-validation only in learning curves.
2. **No hyperparameter search.** Default sklearn parameters throughout.
3. **GloVe mean-pooling.** Arithmetic mean ignores word order.
4. **DistilBERT CLS token.** Mean-pooling across all token positions is often more robust; not explored here.

7 Conclusion and Learnings

This project demonstrated that classical ML performance is fundamentally bounded by input feature quality. The three-tier comparison — sparse features, GloVe, DistilBERT — establishes a hierarchy corresponding directly to representational richness.

The best-performing configuration is RBF SVM on Reduced DistilBERT (PCA) at 90.19% (Macro F1 = 90.09%). PCA on DistilBERT retains 85.89% in 100 components while SVD on TF-IDF retains only 16.19%, making compression viable for embeddings but destructive

for sparse features. The learning curves confirm that DistilBERT is substantially more data-efficient. If you give a classical model a good feature space, it will perform well.